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Children with and without dyscalculia: How mathematics anxiety and executive functions may (or may not) affect mental calculation☆

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ABSTRACT

The aim of the current study was to assess the impact of affective factors, such as mathematics anxiety, and cognitive domain-general factors, like executive functions, on mathematical skills of children with Developmental Dyscalculia (DD) in comparison to non-diagnosed (ND) peers. The study involved 87 children aged between 8 and 15 years old: 39 (17 M) with DD, and 48 (24 M) ND, matched for age, gender, and IQ. Participants completed a mental calculation task, three executive functions tests (inhibition, updating and set-shifting), and a self-report measure of mathematics anxiety. Results suggest higher levels of mathematics anxiety and poorer performance on all executive functions tasks in children with DD. After controlling for children's general anxiety reported by participants' parents, regression analysis revealed that lower levels of mathematics anxiety and better updating skills predicted mental calculation accuracy for the ND participants. However, these factors did not support mathematical performance in children with DD.

Educational relevance and implications statement

A comprehensive assessment of both cognitive abilities and emotional factors is important for identifying potential strengths and weaknesses in addressing mathematical skills. Educators and clinicians should be aware of the negative impact of repeated failures in children with developmental dyscalculia, to prevent the development of negative self-beliefs and learned helplessness. Within the cognitive domain, the ability to retain, manipulate, and update information in memory may represent a critical skill in mathematics, which should be further considered during interventions and in school settings. Individual characteristics, such as anxiety and executive functioning, should also be taken into account when designing interventions aimed at enhancing the mathematical learning process for children with and without mathematical difficulties.

1. Introduction

There is widespread consensus that the effective acquisition of mathematical skills depends on a combination of domain-specific abilities and domain-general aspects (Geary et al., 2017; Gilmore et al., 2018). Domain-specific abilities include basic quantitative skills, such as numeracy, number fact knowledge, counting and calculation abilities, as well as exact numerical representations, such as digit recognition, and access to symbolic and non-symbolic magnitudes (Dowker, 2015; Sasanguie et al., 2012; Schneider et al., 2017). Instead, domain-general skills are related to a complex interplay of underlying cognitive (Caviola et al., 2020; Peng et al., 2016), affective (Barroso et al., 2021; Donolato et al., 2020), and contextual (Demirtas-Zorbaz et al., 2021; Wang et al., 2020) factors that support mathematical learning and performance. Among mathematical skills, arithmetic competence, and in particular mental calculation, has been extensively examined due to its fundamental importance in different academic and everyday situations (e.g., making payments, playing games). Mental calculation has been closely linked to specific affective components, such as mathematics anxiety (Barroso et al., 2021; Caviola et al., 2022), as well as general cognitive abilities, including executive functions (Bull & Lee, 2014; Cragg & Gilmore, 2014). In the current study, the ability of mental calculation has been considered with the aim of investigating the role of both affective factors and cognitive domain-general abilities in typical and atypical mathematical learning.

Regarding affective factors, a specific type of anxiety has been linked to mathematics and is referred to as mathematics anxiety. Alike other

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forms of anxiety, mathematics anxiety encompasses a combination of negative emotions such as tension, discomfort, and fear that individuals experience only when facing or thinking about mathematical tasks, both in academic settings and in daily life (Ashcraft, 2002; Hembree, 1990; Ma & Xu, 2004). Significant small-to-moderate negative associations between mathematics anxiety and mathematics performance have been observed, indicating that students with higher levels of mathematics anxiety tend to have lower levels of mathematics achievement (for reviews and meta-analysis, see Barroso et al., 2021; Carey et al., 2016; Caviola et al., 2022; Hembree, 1990; Ma, 1999; Namkung et al., 2019). However, most studies on this topic are cross-sectional, and the direction of the relationship should be explored through intervention and longitudinal studies. Longitudinal studies with two-time points have yielded mixed outcomes: some suggest that mathematical performance predicts later mathematics anxiety and related attitudes (Ma & Xu, 2004; Song et al., 2021; Vukovic et al., 2013), while others indicate the opposite, with mathematics anxiety leading to changes in performance (Cargnelutti et al., 2017) or even showing reciprocal relationships (Du et al., 2021; Gunderson et al., 2018). Studies examining more complex trajectories, involving more than three time points, generally support reciprocal effects models, where mathematics performance and anxiety influence each other over time (e.g., Aldrup et al., 2020; Lichtenfeld et al., 2023; Pekrun et al., 2017; Szczygieł et al., 2024). Thus, a bidirectional relationship between mathematics anxiety and mathematical skills has been hypothesized (Carey et al., 2016; Cipora et al., 2022; Jansen et al., 2013).

Among domain-general abilities, performance on mental calculation tasks has been found to rely on higher-order cognitive skills, specifically executive functions (Bull & Lee, 2014; Cragg & Gilmore, 2014). This umbrella term refers to a set of top-down mental processes, crucial for planning and monitoring thoughts and behaviours (Diamond, 2013; Miyake et al., 2000). According to Miyake et al. (2000), the three core processes of executive functions are inhibitory control (i.e., the ability to suppress automatic responses), updating (i.e., the capacity to manipulate information in working memory), and cognitive flexibility (i.e., set shifting, or the ability to flexibly switch between tasks). It is worth specifying that in literature, working memory (WM) is described as a comprehensive process that includes both storage and processing, and also encompasses (but is not limited to) updating (Morris & Jones, 1990), as described by Baddeley and Hitch (1974) and Baddeley (2000). While WM is commonly assessed through tasks that evaluate the ability to maintain and process information, WM updating specifically refers to the ability to manipulate and retain the most recently presented information in WM. This ability is often measured using tasks such as the N-back (where participants identify whether an item matches one presented n positions earlier), keeping track (where participants recall the most recent word presented for each category), WM deletion (where participants determine if a probe word belongs to a relevant list), and visuospatial updating (where participants recall the last position of specific shapes in a previously displayed matrix) (Cardillo et al., 2021; Gustavson & Miyake, 2016; Pelegrina et al., 2020; Pelegrina et al., 2024). The importance of studying the association between WM updating and mental calculation lies on the fact that this cognitive process is essential for efficiently manipulating numerical data, continuously updating intermediate results, and managing multiple steps simultaneously during mental calculations. Although many studies have investigated the relationship between WM and mathematics (for reviews see Allen et al., 2019; Friso-van den Bos et al., 2013; Menon, 2016; Peng et al., 2016; Raghubar et al., 2010), there is a knowledge gap on the specific association between WM updating and mathematics (Hu et al., 2023; Lechuga et al., 2016; Lee & Bull, 2016; Pelegrina et al., 2020). Even less has been done to understand the role of inhibitory control and set-shifting. Both inhibition (Bull & Scerif, 2001; Gilmore et al., 2013; St Clair-Thompson & Gathercole, 2006) and shifting ability (Bull & Scerif, 2001; Vosniadou et al., 2018; Živković et al., 2022) seem to predict mathematics outcomes, though set-shifting has found to be strongly associated with intelligence, this latter accounting for a stronger variability on mathematics (effect size: $r = 0.47$, $85\ \%\ CI = 0.41–0.52$) (for a meta-analysis Yeniad et al., 2013). Some studies found inhibition to be more linked to sequential procedures (e.g., counting, word problem solving) rather than calculations (Gilmore et al., 2015; Lan et al., 2011). As regards set-shifting, it seems to be more involved in problem solving rather than mental calculations (Agostino et al., 2010; Van der Sluis et al., 2007). Regarding the crucial role of WM instead, a significant amount of traction has been given to the positive association with mathematical achievement (Caviola et al., 2020; Szücs et al., 2014), with a particular emphasis on its visuospatial component (Geary et al., 2017; Mammarella, Caviola, Giofrè, & Szűcs, 2018; Menon, 2016), regardless the type of mathematical task considered (for a review, see Allen et al., 2019). Previous studies have postulated the significant role of visuospatial WM in mathematics (Allen et al., 2019; Li & Geary, 2013), suggesting it may be more important than verbal WM (Liang et al., 2022; Peng et al., 2018; Szűcs et al., 2014), as mathematical skills heavily rely on the ability to visualize and manipulate spatial representations. Indeed, visuospatial WM appears to be a specific source of vulnerability for mathematical difficulties (Menon, 2016). However, as mentioned earlier, only a limited number of studies examining the relation between WM and mathematics have specifically measured WM updating (Hu et al., 2023; Lechuga et al., 2016; Lee & Bull, 2016; Passolunghi & Pazzaglia, 2005; Pelegrina et al., 2020; St Clair-Thompson & Gathercole, 2006). Even fewer studies have attempted to explore the interplay between both affective and cognitive factors in explaining mathematical attainment (Justicia-Galiano et al., 2017; Mammarella, Caviola, Giofrè, & Borella, 2018; Orbach et al., 2020; Passolunghi et al., 2019; Pellizzoni et al., 2022; Živković et al., 2022). To the best of our knowledge, none of these studies have examined both domains in the context of mathematical difficulties, despite the clinical relevance of this phenomenon.

Developmental Dyscalculia (DD), also known as Mathematical Learning Disability, is defined by severe impairments in the acquisition of mathematical skills. As declared by the international classification systems DSM-5 (American Psychiatric Association, 2013) and ICD-10 (World Health Organization, 1992), individuals with DD consistently perform significantly below the expected level for their age, educational background, and intellectual ability on standardized assessments. These challenges often negatively impact both their academic progress and daily life functioning. Approximately 5–15 % of school-aged children are diagnosed with a Specific Learning Disorder (SLD) (APA, 2013), with demographic studies estimating that 5–6 % of these children experience mathematical learning difficulties (Devine et al., 2013). Regarding affective factors, it is worth noting that some aversive emotional aspects, such as mathematics anxiety, might be intensified in students with established mathematical impairments. Compared to domain-specific and cognitive general aspects, affective factors have been understudied within the context of DD. The few studies on the topic suggest that children with mathematical difficulties experience higher levels of mathematics anxiety compared to those without (Kucian et al., 2018; Lai et al., 2015; Passolunghi, 2011; Rubinsten & Tannock, 2010), given that DD and mathematics anxiety may be dissociated aspects (Devine et al., 2018). Moreover, individual differences in dyscalculic children's self-reports of mathematics anxiety may be further exacerbated by repeated experiences of failure and negative emotional dispositions towards mathematical settings (Hunt & Maloney, 2022; Pekrun, 2006).

A substantial body of literature indicates that specific domain-general cognitive skills, particularly executive functions, are impaired in children with DD (for recent reviews, see Agostini et al., 2022; Mishra & Khan, 2023). Challenges in retaining and simultaneously manipulating verbal, visual and spatial information in WM have been observed in people with DD (Keeler & Swanson, 2001; Kuhn et al., 2016; Mammarella et al., 2015; McDonald & Berg, 2018; Passolunghi & Mammarella, 2012; Szűcs, 2016; Szűcs et al., 2013). Mixed findings have been found when considering inhibition ability in children with DD

(Agostini et al., 2022; Censabella & Noël, 2007; Mammarella, Caviola, Giofrè, & Borella, 2018; Szűcs et al., 2013). Most studies have shown that children with mathematical difficulties struggle to flexibly adapt their responses based on contextual demands (McDonald & Berg, 2018; Van Der Sluis et al., 2004; Willcutt et al., 2013), potentially impacting their capacity to perform complex mathematical calculations that require transitioning between conceptual and procedural knowledge. Although assessing cognitive abilities is recommended as part of the diagnostic process, there is no consensus on which specific skills should be prioritized (Mammarella et al., 2021).

1.1. The present study

In the current study, we assessed mathematical performance using a mental calculation task in children with DD aged between 8 and 15 years old in comparison to non-diagnosed (ND) peers, matched for age, gender and IQ. To be included in the study, children in the DD group had previously received a formal diagnosis of Specific Learning Disorder with significant impairments in mathematics. Additionally, to explore underlying mechanisms of mathematics achievement, we asked participants to complete a self-report measure of mathematics anxiety, as an affective factor related to mathematics. Parents filled in a questionnaire about their children's general anxiety levels, which was useful for controlling the influence of overall tension. To assess domain-general cognitive abilities, we asked participants to complete three executive functions tasks specifically evaluating inhibition, updating and set-shifting abilities.

Our first aim was to examine levels of mathematics anxiety in children with and without DD. It was assumed that children with DD would report higher levels of mathematics anxiety compared to those without DD (Lai et al., 2015; Passolunghi, 2011), due to the well-established negative relationship between mathematics anxiety and mathematical performance (Barroso et al., 2021; Carey et al., 2016; Caviola et al., 2022; Namkung et al., 2019).

The second aim of the present study was to investigate executive functions (inhibition, updating and set-shifting) in a sample of children with DD compared to ND participants. Based on the literature, it was reasonable to hypothesize that children with DD would perform worse on all tasks assessing executive functions compared to ND participants (for a review see Agostini et al., 2022).

The third aim was to understand whether mathematics anxiety and executive functions are associated with mathematical performance, specifically mental calculation, and how these relationships might differ between children with and without DD. It was worth hypothesizing that, after controlling for general anxiety, low levels of mathematics anxiety and strong executive functions could improve mental calculation performance both in children with DD (Lai et al., 2015; Szűcs et al., 2013) and those without DD (Allen et al., 2019; Carey et al., 2016; Caviola et al., 2022; Peng et al., 2016). However, given that children with DD are assumed to have poor working memory (WM) skills (Giofrè & Cornoldi, 2015; Toffalini et al., 2017), this specific weakness might contribute to difficulties with the updating task (Passolunghi & Cornoldi, 2008; Passolunghi & Pazzaglia, 2005; Pelegrina et al., 2015), potentially limiting their ability to improve mental calculation performance. From this perspective, the relationship between executive functions and mathematical performance may not be statistically significant for the DD group.

2. Method

2.1. Participants

The study involved 87 children and adolescents aged between 8 and 15 years old divided into two groups: 39 (17 M) with Developmental Dyscalculia (DD), and 48 (24 M) matched non-diagnosed (ND) participants. Participants of the two groups were matched based on age (within a 6-month range), gender, and IQ (within a 6-point range). The two groups did not differ statistically in chronological age [$F\ (1,\ 85) = 0.64$, $p = .42$, $Cohen's\ d = -0.17$], gender distribution [$X^2 = 0.35$, $df = 1$, $p = .55$], or total IQ [$F\ (1,\ 85) = 0.91$, $p = .34$, $Cohen's\ d = -0.20$]. A summary of the participants' characteristics is shown in Table 1.

The clinical group was assessed at the child and adolescent psychiatric service centers to which they had been referred. All participants in the clinical group had been previously diagnosed with DD or a Specific Learning Disorder with major impairments in mathematics, according to the DSM-5 (APA, 2013) or the ICD-10 (WHO, 1992) criteria. Children and adolescents with DD were identified by assessing their mathematical achievement using standardized batteries (AC-MT-3, Cornoldi et al., 2020; MT-Avanzate-3, Cornoldi et al., 2017), in which they obtained scores >1 SD lower than average (<16th percentile) in at least half of the proposed subtests (e.g., mental calculation, written calculation, transcoding, fact retrieval) (Istituto Superiore di Sanità [ISS], 2022). The DD group of this study has been selected from a larger sample of SLD children, that is part of a wider research project (Lievore, Caviola, & Mammarella, 2024).

The control group was comprised of healthy children without any diagnosis (ND) with no psychiatric, neurological, or neurodevelopmental disorders. They were engaged and examined individually at their respective schools. Participants who were taking medication, or who had other known genetic conditions, a history of neurological diseases, comorbid autism and ADHD, or certified physical and intellectual disabilities were excluded. All children and adolescents were native Italian speakers, and none had any visual or hearing impairments. Participants of both groups were included in this study only if they achieved a standard score of 80 or more for the abbreviated version of IQ on the Wechsler Intelligence Scales (WISC IV; Wechsler, 2003).

2.2. Materials

2.2.1. Mental calculation

This task, adapted from Caviola et al. (2016); Caviola et al. (2018), consisted of 60 multiple-choices trials divided into two blocks of 30 trials each: the first block included two-digit additions without carrying, the second two-digit subtractions without borrowing. Participants had to choose the correct answer between the three alternative choices (the correct answer, the correct answer plus or minus 1, and the correct answer plus or minus 10). The operations to be solved appeared at the top of the screen with the three answer options placed horizontally underneath. The order of the possible answers was counterbalanced to prevent children from spotting a response pattern. Participants had to press a keyboard key based on the position on the screen of the answer they wanted to select ("z" for the left choice, "v" for the one in the middle, and "m" for the right one). Participants had to solve each operation within the time limit of 10 s, as shown by the presence of a count-down clock on the screen. Before starting the real task, participants had the chance to solve three practise trials with feedback. The total accuracy was considered in the statistical analyses of the present study.

Table 1
Descriptive statistics and statistical comparisons in children and adolescents with Developmental Dyscalculia (DD) and without any diagnosis (ND).

Screening variables	**DD (*n* = 39)**		**ND (*n* = 48)**		***F* (1, 85)**	***p***	***Cohen's d***
Gender M:F	17:22		24:24				
	M	**SD**	**M**	**SD**			
Age (years)	11.28	1.79	11.62	2.12	0.64	0.42	−0.17
Age range	8–15		8–14				
IQ	102.92	10.14	104.94	9.52	0.91	0.34	−0.20
IQ range	83–126		83–120				

Note: SD: standard deviation; IQ: intelligence quotient.

Cronbach's α = 0.91 (C.I. = 0.88–0.94).

2.2.2. Questionnaires

2.2.2.1. General anxiety. The parents' version of the Multidimensional Anxiety Scale for Children (MASC-2; March, 2012; Italian version, Paloscia et al., 2017) was administered. It is a 50-item questionnaire that evaluates the presence of anxiety on a rating scale that goes from 0 ("never") to 3 ("often"). It includes a total score, and six subscales (i.e., separation anxiety, generalized anxiety disorder, social anxiety, obsessions and compulsions, physical symptoms, and harm avoidance). Raw scores are converted into T scores by using normative data that considers the child's age and gender. This tool also features an Inconsistency Index that identifies possible unreliable ratings by comparing scores on eight item-pairs with the highest bivariate inter-item correlations from the development sample. In this study, the total T score has been considered.

Cronbach's α =0.89 (C.I. = 0.88–0.90).

2.2.2.2. Mathematics anxiety. Children were requested to fill in the Abbreviated Math Anxiety Scale (AMAS; Hopko et al., 2003; Caviola et al., 2017), which involves nine Likert-type items ranging from 1 ("No bad feelings") to 5 ("Very bad feelings") related to feelings and level of fear towards mathematics. Examples of items are: "Having to solve many difficult mathematical problems for homework due the next class lesson" or "Starting a new topic in mathematics". Higher scores on the scale indicate higher levels of MA. In this study, the total raw score was considered. We administered the Italian version of the AMAS (Caviola et al., 2017).

Cronbach's α = 0.86 (C.I. = 0.83–0.88).

2.2.3. Executive functions

The tripartite model of executive functions by Miyake (Miyake et al., 2000) was employed to evaluate executive functioning in our sample. This model encompasses three components: inhibition of impulsive reactions ("Inhibition"), updating and monitoring ("Updating"), and mental set shifting or flexibility ("Set-shifting").

2.2.3.1. Inhibition. Inhibitory control was assessed with a computerized go/no-go task (Lievore, Cardillo, & Mammarella, 2024; adapted from Christ et al., 2007). The task included 120 trials, composed by two blocks (60 + 60) with a break between them. On each trial, one of four stimuli (blue, red, yellow, and green dots) was centrally displayed in a computer monitor. In the initial block, participants were asked to press the spacebar as quickly as possible when a blue dot appeared (target; go trials) and to give no response when a dot in any other colour showed (non-target; no-go trials). In the subsequent block, children were instructed to quickly press the spacebar whenever a dot of any colour except blue materialized (referred to as target; go trials), and to abstain from reacting when a blue dot was presented (classified as non-target; no-go trials). Performance on the go trials corresponds to a measure of attention, whereas errors on the no-go trials to a measure of inhibition. Within each block, stimuli were produced in a random order and targets were presented on a minority of trials (25 %). After an intertrial interval of 2000 ms, a new trial was presented. At the beginning of the task, following the delivery of instructions, eight practice trials were introduced, during which children received feedback regarding their responses ("correct", "incorrect", and "too slow" when the response was not given within 2.000 ms). Errors on the no-go trials (commission errors = number of reactions to the non-target stimuli) were considered as a measure of inhibition.

Cronbach's α (for both go and no-go trials) = 0.94 (C.I. = 0.92–0.96).

2.2.3.2. Updating. This computerized task, modified from the work of Cardillo et al. (2021) and Crisci et al. (2021), was employed to evaluate the ability for visually and spatially refreshing information within the working memory system. Participants were shown a 4 × 4 blank matrix on the computer screen. A total of eight sequences of various shapes (i.e., triangle, circle, star, square, rhombus, pentagon) appeared in the matrix. Participants were asked to recall the last position of specific shapes in the matrix previously displayed on the screen. The quantity of shapes to be recalled varied between 2 and 5, contingent on the difficulty level of the task. Sequences of 6, 8, 10, or 12 shapes were used, and there were two trials for each sequence length. Before starting the experiment, participants were exposed to three practice trails. Accuracy was calculated as a proportion of items' last positions correctly recalled out of the total positions to remember (Giofrè & Mammarella, 2014).

Cronbach's α = 0.80 (C.I. = 0.75–0.83).

2.2.3.3. Set-shifting. A computerized version of the Wisconsin Card Sort Test (WCST; Heaton et al., 1993; Lievore, Cardillo, & Mammarella, 2024) has been designed from the original edition. Every trial consisted of four stimulus cards and one response card, with the participant's objective being to accurately associate the response card with one of the stimulus cards based on one of three grouping principles. Following each trial, participants were provided with feedback regarding the accuracy of their answers. The stimulus cards were displayed in a row at the upper part of the computer screen, while the response cards appeared one at a time at the lower section of the screen. The four stimulus cards displayed the following symbols: (a) one red triangle, (b) two green stars, (c) three yellow crosses, and (d) four blue circles. There were two decks of 64 response cards, each containing from one to four identical symbols (triangles, stars, crosses, or circles) of a particular colour (red, green, yellow, or blue). The first correct sorting principle was colour, followed by symbol and then number. The participant had to correctly match the stimulus and the response cards 10 times consecutively within each criterion. The test concluded either when the participant successfully matched the response cards according to six criteria (twice for each sorting principle), or when they employed two sets of 64 response cards (totalling 128 cards). The analysis of the present study involved using the total count of perseverative errors, which were computed as proportions relative to the total number of administered trials. A perseverative error is defined as a failure to shift category after receiving negative feedback from the previous trial.

Generalisability coefficient ranges between 0.39 and 0.72 (Heaton et al., 1993).

2.3. Procedure

The study was approved by the ethics review board of the Authors' institution and adheres to APA ethical standards. After obtaining the written consent of children's parents to their participation in the study, participants were tested individually in a quiet room at specialized centers (DD) or at school (ND). The sessions lasted approximately 30 min each. Participants in the ND group underwent a screening phase followed by two experimental sessions. Instead, we chose participants with DD from a larger group with SLD who part of a broader study had already been. We reached out to these participants and their families to administer additional tests on executive functions and mental calculation. Since the DD group had already completed the screening phase and the AMAS as part of the prior study, they only participated in the two experimental sessions. Aside from these differences, the test administration procedures were identical for both groups. In the screening phase (for the ND group), the IQ was calculated and only participants who scored above 80 were included. The first experimental session included the mental calculation task, the inhibition task, and the AMAS. The second session comprised the set-shifting and the updating tasks. The tasks in the experimental sessions were presented in a counterbalanced order. The experimenter provided instructions for each task, letting the participant practice before starting the experiment. The computerized tasks (mental calculation and executive functions tasks) were created and administered using PsychoPy3 (Peirce et al., 2019), and a laptop computer with a 15-in. LCD screen. At the same time, parents completed the MASC-2 for assessing the child general anxiety.

2.4. Statistical approach

Before performing the analyses, an outlier analysis was conducted through visual inspection, identifying a few outlier participants, all from the DD group ($N = 7$; mental calculation = 5, inhibition = 1, set-shifting task = 1). Sensitivity analyses indicated that the overall pattern of results remained consistent regardless of whether these outliers were included or excluded. Consequently, they were retained to avoid misrepresenting the data, as their presence was not attributed to data entry or measurement errors (Bakker & Wicherts, 2014).

First, a series of univariate ANOVAs were performed to estimate differences between the two groups in the measures of interest. Effect sizes were also computed using Cohen's d, which expresses the effect size of the pairwise comparisons between the groups. Moreover, the Benjamini-Hochberg False Discovery Rate (FDR) method was used to correct for multiple comparisons to detect the likelihood of type I errors. Due to the non-normal distribution of the variables, Spearman correlation analyses, divided by group (DD and ND), are presented in Table 3. Group-specific differences (DD, ND) between boys and girls across the variables, are reported in Table S1 (Supplementary material).

Second, a series of regression models using a hierarchical approach was performed to assess the effect of the measured variables (general anxiety, mathematics anxiety, inhibition, updating, set-shifting) on the dependent variable (total accuracy on the mental calculation task). In the first model, we included the general anxiety as a control variable; in the second model, we added the group; in the third, the mathematics anxiety; in the fourth, the executive functions (inhibition, updating, and set-shifting) measures were included. The interactive effect of Group (i.e., DD, ND) with MA, inhibition, updating, and set-shifting was added in the fifth and last model. The residuals of the last regression model were found to be normally distributed, as indicated by the Shapiro-Wilk test ($W = 0.98$, $p = 0.18$) and confirmed by visual inspection of the Q-Q plot. There were two missing data points related to the administration of the computerized WCST (set-shifting measure), during which there was an error causing the data not to be saved correctly. The regression models were automatically computed excluding those observations.

The five models were compared using ANOVAs; the more complex model was compared with the simpler one to understand which one was significantly better at capturing the data. The best model was then selected using information-theoretic (I-T) approaches (Burnham et al., 2011), considering the Akaike information criterion (AIC), the R^2 and the adjusted R^2 (Adj R^2). The AIC is an estimator of prediction error and therefore of the relative quality of statistical models for a given set of data: a lower AIC indicates a better model. R^2 assesses the proportion of variance in the dependent variable explained by the model, with higher values indicating better explanatory power. The Adj R^2 is a modified version of R^2 that adjusts for the number of predictors in a regression model and allows for comparing models with a different number of predictors (Miles, 2005). Moreover, the log-likelihood (logLik) and the RMSE (Root Mean Square Error) were also calculated: the first reflects the probability of the data given the model, where higher values suggest a better fit; the second measures the average deviation of predictions from observed values, indicating model accuracy.

Data were analyzed using R version 4.2.1 (R Core Team, 2022). The "stats" package was used to run univariate ANOVAs and adjust p-values for multiple comparisons, the "lme4" package was used to run the regression models and the AIC (Bates et al., 2015), and the "ggplot2" package to obtain the graphical effects (Wickham, 2016).

3. Results

3.1. Preliminary analysis

Table 2 displays descriptive statistics and statistical comparisons between the two groups in all considered measures. Effect sizes of the pairwise comparisons between the groups expressed with *Cohen's d* are also reported. Corrections for multiple comparisons using the FDR were applied, and the p-values did not undergo significant changes. Table 2 also presents each variable's skewness and minimum/maximum values, separated by group (DD, ND). Table 3 includes Spearman's correlations between the variables, divided by group (DD, ND).

The DD group achieved an overall lower performance in the mental calculation task (total accuracy) as compared to the ND group, $F(1, 85) = 121.28$, $p < 0.001$, *Cohen's d* = −2.38; DD = 0.37 (0.13), ND = 0.76

Table 2
Descriptive statistics and statistical comparisons by group for all considered measures in children and adolescents with Developmental Dyscalculia (DD) and without any diagnosis (ND).

Measures	DD (n = 39)		ND (n = 48)		*F* (1, 85)	*p*	*Cohen's d*	*FDR*
	M	SD	M	SD				
Mental calculation	0.37	0.13	0.76	0.18	121.28	**<0.001**	−2.38	**<0.001**
Min-max	0.10 – 0.68		0.32 – 0.98					
Skewness	−0.05 (0.38)		−0.88 (0.34)					
Questionnaires								
General anxiety	62.72	12.00	55.21	10.69	9.51	**0.003**	0.66	**0.004**
Min-max	40 – 80		40 – 82					
Skewness	0.18 (0.38)		0.18 (0.34)					
Mathematics anxiety	25.92	7.27	22.67	5.58	5.58	**0.02**	0.51	**0.02**
Min-max	9 – 38		12 – 34					
Skewness	−0.75 (0.38)		0.05 (0.34)					
Executive functions								
Inhibition	5.00	3.43	2.12	1.94	24.26	**<0.001**	1.06	**<0.001**
Min-max	1 – 15		0 – 8					
Skewness	0.87 (0.38)		1.54 (0.34)					
Updating	0.57	0.24	0.73	0.15	14.28	**<0.001**	−0.81	**<0.001**
Min-max	0.05 – 0.88		0.27 – 0.95					
Skewness	−0.62 (0.38)		−1.06 (0.34)					
Set-shifting	15.95	6.73	12.78	6.88	4.54	**0.04**	0.46	**0.04**
Min-max	6.25 – 35.16		3.00 – 37.50					
Skewness	1.05 (0.38)		1.52 (0.35)					

Note: Mental calculation = proportion of total accuracy; General anxiety = T score on the total scale of the MASC-2; Mathematics anxiety = total raw score on the AMAS; Inhibition = total errors on the no-go trials; Updating = proportion of correct responses; Set-shifting = percentage of perseverative errors on the WCST task; FDR = False Discovery Rate.

Table 3
Spearman's correlational analyses on measured variables divided by group (DD in the lower diagonal, ND in the upper diagonal).

Variables	1.	2.	3.	4.	5.	6.	7.	8.
1. Age	-	-0.17	0.29*	-0.26	-0.15	0.09	0.41*	0.18
2. IQ	0.15	-	0.37**	0.04	-0.40**	0.23	0.28	-0.08
3. Mental calculation	0.34*	0.23	-	0.01	-0.46**	-0.02	0.51**	0.14
4. General anxiety	-0.09	-0.07	-0.29	-	0.12	-0.14	0.04	-0.11
5. Mathematics anxiety	0.04	-0.31	-0.11	0.18	-	30.0	-0.26	-0.19
6. Inhibition	-0.19	-0.07	-0.07	0.05	0.11	-	-0.11	-0.03
7. Updating	0.35*	0.44**	0.08	0.21	-0.19	-0.15	-	0.02
8. Set-shifting	-0.24	-0.16	-0.17	0.14	0.22	-0.06	-0.14	-

Note: IQ, intelligence quotient. $p < .05$*, $p < .001$**.

(0.18). Figure S1 (Supplementary materials) shows the distribution of the accuracy scores on the mental calculation task divided per group (DD, ND) with individual observations.

Concerning the questionnaires, participants with DD had greater levels of general anxiety (as reported by their parents), $F(1, 85) = 9.51$, $p = 0.003$, *Cohen's* $d = 0.66$, and of self-reported mathematics anxiety, $F(1, 85) = 5.57$, $p = 0.02$, *Cohen's* $d = 0.51$, than the ND group.

As regards the executive functions tasks, the DD group performed significantly worse than the ND group in all tasks. Statistically, participants with DD made more errors in the inhibition task (no-go condition), $F(1, 85) = 24.25$, $p < 0.001$, *Cohen's* $d = 1.06$, performed worse in the updating task (total accuracy), $F(1, 85) = 14.24$, $p < 0.001$, *Cohen's* $d = -0.81$, and made more perseverative errors in the set-shifting task (WCST), $F(1, 83) = 4.54$, $p = 0.04$, *Cohen's* $d = 0.46$, than the ND group.

3.2. Hierarchical regression analysis

A series of linear regressions (see Table 4) were tested to investigate the association between the performance on the mental calculation task and the hypothesized predictors (i.e., general anxiety, mathematics anxiety, inhibition, updating and set-shifting). The interactive effect of Group with other predictors was also considered. Models' comparison is also shown in Table 5 with the results of the ANOVAs between models, and metrics of fit (AIC, logLik, RMSE, R^2, adjusted R^2).

Our model fitting procedure revealed that the best-fitting model was Model 5: Mental calculation (total accuracy) ~ General anxiety + Group + Mathematics anxiety + Inhibition + Updating + Set-shifting + Group*Mathematics anxiety + Group*Inhibition + Group*Updating + Group*Set-shifting ($F = 3.22$, $df = 4$, p = 0.02; fit indexes: $AIC = -78.48$, $\Delta°AIC = 81.89$, $logLik = 51.24$, $RMSE = 0.13$, $R^2 = 0.72$, $adjusted\ R^2 =$

Table 4
Hierarchical regression analysis with mental calculation (total accuracy) as dependent variable, and all the other measured variables and their interactions with group (DD, ND) as predictors.

Regression models	*Standardized beta*	*Estimate coefficient*	*SE*	*t*	*p*
Model 1					
General anxiety	**−0.29**	**−0.006**	**0.002**	**−2.89**	**0.005**
Model 2					
General anxiety	−0.06	−0.001	0.001	−0.85	0.39
Group	**0.74**	**0.37**	**0.04**	**10.16**	**<0.001**
Model 3					
General anxiety	−0.02	−0.001	0.001	−0.33	0.74
Group	**0.72**	**0.36**	**0.04**	**9.86**	**<0.001**
Mathematics anxiety	**−0.17**	**−0.007**	**0.003**	**−2.42**	**0.02**
Model 4					
General anxiety	−0.07	−0.001	0.001	−1.05	0.32
Group	**0.66**	**0.33**	**0.04**	**8.04**	**<0.001**
Mathematics anxiety	**−0.16**	**−0.005**	**0.003**	**−2.19**	**0.03**
Inhibition	0.01	0.001	0.006	0.23	0.81
Updating	**0.16**	**0.18**	**0.09**	**2.02**	**0.04**
Set-shifting	−0.03	−0.001	0.002	−0.40	0.69
Model 5					
General anxiety	−0.06	−0.001	0.001	−0.92	0.36
Group	0.49	0.25	0.22	1.14	0.26
Mathematics anxiety	−0.01	−0.001	0.003	−0.18	0.86
Inhibition	0.003	0.06	0.10	0.57	0.57
Updating	0.05	0.001	0.007	0.04	0.97
Set-shifting	−0.13	−0.004	0.003	−1.31	0.19
Group * Mathematics anxiety	**−0.55**	**−0.01**	**0.005**	**−2.20**	**0.03**
Group * Inhibition	0.01	0.002	0.01	0.16	0.87
Group * Updating	**0.65**	**0.43**	**0.19**	**2.30**	**0.02**
Group * Set-shifting	0.11	0.003	0.005	0.73	0.46

Table 5
Models' comparison.

Models	*F*	*df*	*p*	*AIC*	*logLik*	*RMSE*	R^2	*Adj* R^2
Model 1	–	–	–	3.41	1.29	0.24	0.09	0.08
Model 2	138.79	1	<0.001	−64.29	36.14	0.16	0.59	0.58
Model 3	7.65	1	0.007	−68.23	39.11	0.15	0.62	0.60
Model 4	1.61	3	0.19	−72.83	44.42	0.14	0.67	0.65
Model 5	**3.22**	**4**	**0.02**	**−78.48**	**51.24**	**0.13**	**0.72**	**0.68**

Note: AIC, Akaike Information Criterion; logLik, log-likelihood; RMSE, Root Mean Square Error. The lower the AIC and the RMSE, the higher the logLik and the R^2, the better the model.

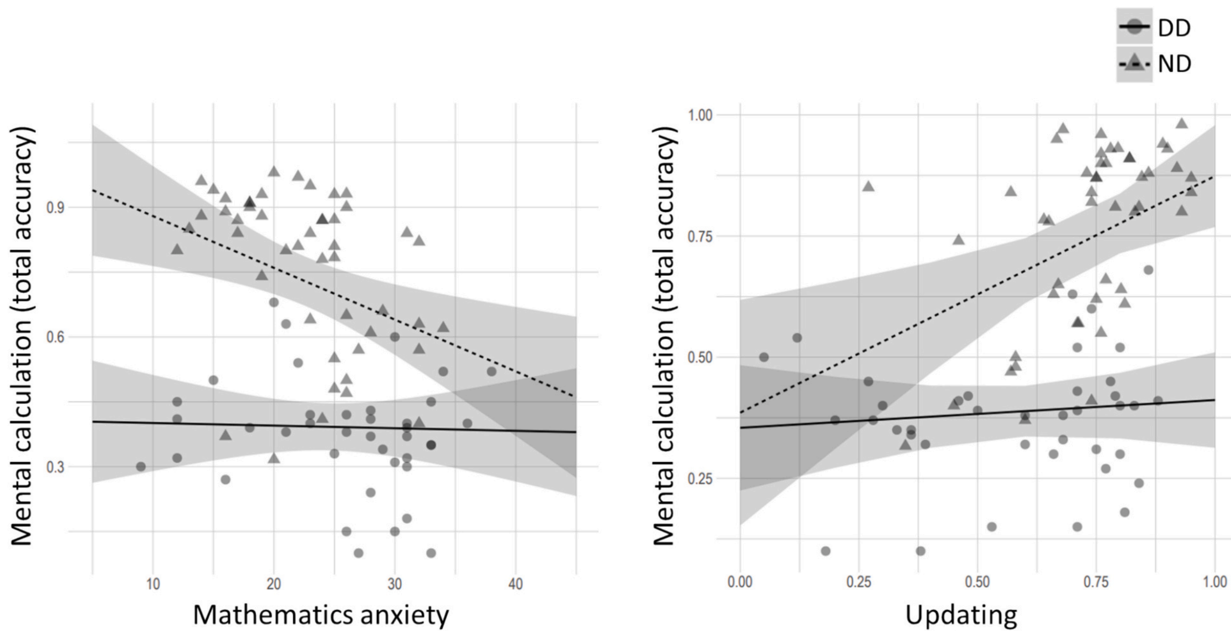

Fig. 1. Significant interaction effect of the Group*Mathematics anxiety and Group*Updating, with Mental calculation (total accuracy) as dependent variable. Error bands represent 95 % confidence intervals.

0.68). Two interaction effects were found to be statistically significant: between Group and Mathematics anxiety, $t = -2.20$, $p = 0.03$; and between Group and Updating, $t = 2.30$, $p = 0.02$. More specifically, as shown in Fig. 1, lower levels of mathematics anxiety and higher scores in the updating task coincided with a better performance in the mental calculation task in the ND group, but not in the DD group. No other statistically significant main or interactive effects emerged in the best-fitting model.

4. Discussion

The main aim of the current study was to evaluate factors associated with mathematical skills, specifically mental calculation, of children with DD in comparison to ND peers, matched for age, gender and IQ. In particular, we investigated the role of affective factors and cognitive domain-general abilities, respectively mathematics anxiety and executive functions, in predicting mathematical skills of children with and without DD.

The study introduces several innovative elements, which we consider strengths, that have never been addressed in previous studies. First, we employed a case-control design to identify condition-specific characteristics by comparing children meeting criteria for a DD diagnosis to a control ND group. Significantly, the children in the DD group had already received a formal clinical diagnosis following DSM-5 (APA, 2013) or ICD-10 (WHO, 1992) criteria, specifically highlighting severe mathematical impairments. This approach is not a negligible aspect, as studies on the topic often identify SLD cases within the general population based on some screening measures, rather than through formal diagnoses (Mammarella et al., 2021). Second, computerized tasks (i.e., mental calculation and executive functions) have been used, providing a higher reliability of the performance than traditional assessments. These tasks allow for precise control over task presentation, minimizing potential sources of bias and variability in data. Third, the research uniquely examines executive functioning as a multicomponent psychological process (Diamond, 2013; Miyake et al., 2000), exploring how each individual component influence children's mental calculation abilities. Moreover, this is the first study to explore the specific contribution of working memory updating, rather than general working memory, to mathematical performance considering a sample of diagnosed dyscalculic children. Finally, the study incorporates general anxiety as a covariate in the first step of the regression model, ensuring that a constant state of tension would not represent a confounding variable in the relationship between the primary predictors and the dependent variable (Carey et al., 2017), a consideration overlooked in earlier research.

Our findings confirmed the first hypothesis regarding levels of mathematics anxiety in children with and without DD. As previously observed (Kucian et al., 2018; Lai et al., 2015; Passolunghi, 2011; Rubinsten & Tannock, 2010), the DD group reported higher mean levels of mathematics anxiety compared to the ND group. This underscores how affective factors can contribute to the discomfort experienced by diagnosed children at school. Repetitive failures in mathematical tasks might be a risk factor for developing negative emotions, thoughts, and worries (Carey et al., 2017; Hunt & Maloney, 2022; Pekrun, 2006), potentially leading to the avoidance of math-related activities, including in real-life contexts.

Our second hypothesis regarding executive functions in children with and without DD has also been confirmed. Statistical comparisons show significant differences between the two groups in terms of inhibition, updating and set-shifting. Concerning inhibitory mechanisms, children with DD made a greater number of commission errors in the go-no go task, showing a higher probability of failure to inhibit a response that should be withheld. In this regard, inhibitory control allows students to stop unsuccessful problem-solving strategies and choose more efficient ones, by also preventing impulsive errors, thus children with DD might be disadvantaged by weak inhibitory mechanisms (Agostini et al., 2022; Mammarella, Caviola, Giofrè, & Borella, 2018; Szűcs et al., 2013). As regards updating skills, children with DD seem to be characterized by the inability to maintaining, manipulating and/or updating the visual and spatial information in WM (Keeler & Swanson, 2001; Kuhn et al., 2016; Mammarella et al., 2015; Mammarella, Caviola, Giofrè, and Borella, 2018; McDonald & Berg, 2018; Passolunghi & Mammarella, 2012; Szűcs et al., 2013; Szűcs, 2016), which represents a critical skill in academic learning, especially in mathematics (Passolunghi & Pazzaglia, 2005; St Clair-Thompson & Gathercole, 2006). Children with DD also made a significant higher number of perseverative errors in the WCST, as an indicator of difficulties with mental flexibility, set-shifting and planning (McDonald & Berg, 2018; Van Der Sluis et al., 2004; Willcutt et al., 2013), which could contribute to worse scholastic learning and achievement (Bull & Scerif, 2001).

Besides the differences between the two groups in the measures of interest, our third (and main) goal was to investigate whether mathematics anxiety and executive functions might be associated with mathematical performance (specifically mental calculation accuracy), and how these relationships might differ among the groups with and without DD. After controlling for general anxiety, our best-fitting model revealed two significant interaction effects, predicting the mental calculation performance: the first between group and mathematics anxiety, and the second between group and updating skills. Results indicated that the relationship between these predictors and mental calculation vary depending on the group membership (DD, ND).

In agreement with our initial hypotheses, a negative relationship between the level of mathematics anxiety and mental calculation accuracy occurred in the ND group. Statistical comparisons showed that ND children reported lower levels of mathematics anxiety as compared to those who experience mathematical challenges: this could suggest a good affective disposition towards mathematics, probably as a consequence of the lack of difficulties in solving operations. Therefore, lower levels of mathematics anxiety might be a supporting factor for mathematical achievement in ND children (Carey et al., 2016; Caviola et al., 2022). Instead, no linear relationship has been found between mathematics anxiety and mental calculation skills in children diagnosed with DD. Indeed, though the significant difference with the ND group, children with DD exhibited greater variability in the levels of reported mathematics anxiety. On one hand, there are individuals with DD who experience mathematics anxiety and emotional problems, but on the other hand, there are also students with DD not suffering from mathematics anxiety. This confirms that mathematics anxiety and DD can be dissociated, as proposed by Devine et al. (2018). In fact, children with DD might be characterized by secondary symptoms, such as a reduced motivation towards mathematics, low perception of competence and self-efficacy (Novita, 2016; Riddick, 2009; Terras et al., 2009), not caring of their mathematical flaws, thus feeling little mathematics anxiety. Similarly, children with DD might report high levels of mathematics anxiety regardless of their ability of solving mental calculations, and despite the possible improvements, possibly because of learned helplessness (Peard, 2010; Rubinsten & Tannock, 2010). All things considered, the large variability of mathematics anxiety levels in DD may prevent the linear relationship with calculation abilities found in the ND group.

In addition, a positive relationship between updating skills and mental calculation accuracy appeared in the ND group. Our regression analysis suggests that updating skills, among all the executive functions considered, might be protective and supportive of mental calculation performance in the ND group (Hu et al., 2023; Lee & Bull, 2016; Passolunghi & Pazzaglia, 2005). As a fact, mental calculation frequently requires students to retain information in their memory and continuously update partial results to arrive at the final one. Intact cognitive abilities might strengthen mathematical achievement, so children without mathematical impairments may perform more effectively due to these sustaining characteristics (Friso-van den Bos et al., 2013). Like the previous result, no linear relationship was found between updating skills and mental calculation performance in children diagnosed with DD. It is worth noting that children with DD exhibited lower average performance on both tasks, which may have restricted the range of scores. This limited distribution could, in turn, have reduced the likelihood of identifying a significant correlation through statistical analysis. In addition, we assumed children with DD to have a poor WM profile (Giofrè & Cornoldi, 2015; Toffalini et al., 2017). Thus, the diminished WM functioning (including updating) of children diagnosed with DD might be incapable of providing performance support. On the other hand, it could be possible that preserved updating skills, whether present, are not sufficient to overcome the core mathematics difficulties experienced by participants with DD.

Our results did not provide sufficient evidence to establish the role of inhibition and set-shifting in influencing mathematics performance within our sample. As for inhibition, this skill involves suppressing irrelevant information and impulses. While important for focusing on tasks and avoiding distractions, it may not directly contribute to the step-by-step processing and updating required in mental calculations. Indeed, some studies found a stronger role of inhibition in executing sequential procedures (e.g., counting, word problem solving) rather than recalling number facts (Gilmore et al., 2015; Lan et al., 2011; Lemaire & Lecacheur, 2011; Winegar, 2013). Regarding set-shifting, or the ability to switch between mental sets, it is crucial for flexibility particularly in tasks that require changing approaches and alternating between different strategies, for example complex problem solving (Agostino et al., 2010; Bull et al., 2008; Mayes et al., 2009; Van der Sluis et al., 2007). However, it may be less directly involved in the continuous integration required for mental calculations.

The present study is not without limitations, which could help to detect future directions of our study. First, mathematics anxiety has been investigated by asking children to fill out a trait-like self-report measure, implicitly legitimizing the idea that mathematics anxiety is a permanent trait. However, questionnaires could be influenced by subjective beliefs and reappraisal thoughts related to past experiences, thus it could be important to assess state mathematics anxiety in real-time assessments (Lievore, Caviola, & Mammarella, 2024; Mammarella et al., 2023), given the clear trait-state discrepancy (Fernández-Blanco et al., 2024; Pelegrina et al., 2024). Previous studies revealed that state mathematics anxiety may play a more significant role in the interplay between anxiety, performance, and executive functions compared to trait anxiety (Orbach et al., 2019; Orbach et al., 2020; Sorvo et al., 2017). Second, we considered mathematics anxiety as a unitary construct, setting aside its multicomponent nature, since it comprises a combination of negative worries, arousing emotions, and unpleasant self-beliefs. Future studies should assess the concordance between state and trait mathematics anxiety, and its different cognitive, emotional, and physiological components (Ashcraft, 2019; Cipora et al., 2022; Ho et al., 2000). Third, regarding domain-general cognitive abilities, our study focused on the tripartite model of executive functions (Miyake et al., 2000), not considering other crucial skills of the mathematical processing network that may play a role in mathematical achievement, for example phonological and spatial abilities (Szűcs et al., 2014). Future research should also consider achievement emotions, values and perceived control towards mathematics (Forsblom et al., 2022), to gain a comprehensive idea of underlying (and supporting) mechanisms of mathematical achievement in children with and without DD.

Future research on the relationship between mathematics anxiety, executive functions, and mathematical performance could build on these findings discussed by addressing key methodological and theoretical gaps. Although previous studies in the literature (e.g., Lehto et al., 2003; Miyake et al., 2000) recommend assessing executive functions through multiple tests targeting the same factor (inhibition, WM updating, set-shifting), we were unable to follow this approach to prevent test fatigue in our sample, which also included special conditions. Therefore, future research should incorporate a broader range of tasks to achieve a better understanding into how these skills may influence mathematical abilities. Moreover, longitudinal designs involving mathematics anxiety, executive functions, and mathematical performance should be implemented to ensure accurate temporal sequencing (Pellizzoni et al., 2022; Szczygieł, 2021). This approach would allow researchers to use mediation models, offering clearer insights into causal relationships when exploring the interplay between these variables (Cain et al., 2018; Fairchild & McQuillin, 2010). Moreover, real-time or simultaneous assessments of both mathematics anxiety and working memory during mathematical tasks could provide valuable insights into how anxiety dynamically affects cognitive resources, as suggested by attentional control theory (Eysenck et al., 2007), potentially revealing more meaningful mediation effects or other temporal dynamics (Finell et al., 2022). Future research could also explore whether individual differences in working memory or other executive functions moderate the impact of mathematics anxiety on mathematical performance (Korem et al., 2022; Orbach et al., 2020). These types of measurements also enable the development of more complex models that combine both mediation and moderation analyses, offering a deeper understanding of how various executive functions jointly influence the relationship between mathematics anxiety and performance. For example, testing moderated mediation models might reveal how the indirect effects of anxiety on performance, via working memory, vary depending on different levels of executive functioning.

Even with the above-mentioned limitations, our findings could have important clinical and educational implications. Educators and clinicians should bear in mind that underlying affective and cognitive factors could contribute to influence the mathematical performance of children with and without DD. As a consequence, a global assessment of both abilities and emotional aspects may be important to identify potential strengths and weaknesses of children profile. For example, the identification and modification of mathematics anxiety, especially of negative thoughts and self-beliefs, could represent a starting point for addressing learning difficulties, even before working on the use of compensatory strategies to reinforce mathematical concepts. Similarly, the assessment of executive functions and WM could represent a valuable source of information useful to tailor the interventions based on the individual characteristics.

In conclusion, the present study confirmed previous findings of higher levels of mathematics anxiety and reduced executive functions in children with DD compared to those without the diagnosis. More interestingly, our results suggest the predictive role of lower levels of mathematics anxiety and stronger updating skills on mental calculation for children without DD. However, these factors do not appear to support mathematical performance in children with DD.

CRediT authorship contribution statement

Rachele Lievore: Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sara Caviola:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Irene C. Mammarella:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Ethics approval statement

The study was approved by the Ethics Committee on Psychology Research at the University of Padova.

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Declaration of competing interest

The authors declare no competing interests.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.lindif.2025.102693.

Data availability

The data that support the findings of this study are available on request from the corresponding authors. The data are not publicly available due to privacy restrictions.

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